

Python URL Shortener

Internship Project Presentation

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1. Project Overview

- **The Problem:** Long, complex URLs are difficult to share, prone to formatting errors, and look unprofessional in documents.
- **The Solution:** A lightweight, offline-first Python desktop application that compresses URLs into unique 6-character codes.
- **Privacy Focused:** Unlike Bitly or TinyURL, this application runs locally, ensuring user data and link history remain private.

2. Technology Stack

The project was developed entirely using Python's standard libraries, requiring no external dependencies:

- **Tkinter:** Used to design the Event-Driven Graphical User Interface (GUI).
- **JSON Module:** Facilitated data serialization to permanently store the URL mapping on the local hard drive.
- **String & Random:** Handled the cryptographic generation of the alphanumeric short codes.
- **OS Module:** Managed file system checks to ensure the database loaded correctly without crashing.

3. Key Application Features

- **Collision Prevention:** An algorithmic loop guarantees that no two short codes are ever duplicated in the database.
- **Input Validation:** Strictly enforces standard `http://` or `https://` prefixes and blocks empty submissions.
- **Clipboard Integration:** Users can extract generated links instantly via a dedicated "Copy to Clipboard" button.
- **Graceful Error Handling:** Replaced terminal crashes with intuitive pop-up warning messages for the end-user.

4. Learnings & Future Scope

- **Internship Outcomes:** Gained practical exposure to the Software Development Life Cycle (SDLC), transitioning a simple terminal script into a deployable GUI app over 4 weeks.
- **Future Enhancements:**
 - Migrate local JSON storage to a cloud database (e.g., Firebase).
 - Refactor the Tkinter frontend into a Flask web application.
 - Implement analytics to track how many times a link is clicked.

Thank You!